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Wafer-scale miniaturized slow-light silicon modulators for 400 Gbps interconnects

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Silicon Mach–Zehnder modulators (MZMs) have become indispensable in optical interconnects for data centers and AI clusters due to their robustness and scalability. However, its working principle makes it challenging for silicon MZMs to simultaneously achieve the bandwidth necessary to support 400 Gbps/lambda PAM-4 transmission while maintaining practical modulation efficiency. Here, we present an O-band slow-light silicon MZM leveraging photonic crystal (PhC) waveguides, fabricated on a 300 mm silicon photonics platform. By exploiting the slow-light effect to enhance light–matter interaction, the device achieves a compact footprint of 500 μm, a low half-wave voltage-length product ($V_\pi \cdot L$) of 0.65 V · cm, and a high median electro-optic bandwidth of 94.7 GHz. Operated with a 2.7 V_{pp} driving voltage, the modulator demonstrates its potential for high-speed, high-density optical interconnects by supporting optical transmission rates of 300 Gbps/λ (135 Gbaud PAM-4) and 400 Gbps/λ (135 Gbaud PAM-8) with a low power consumption of 15.0 fJ/bit.

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1. INTRODUCTION

Driven by the rapid proliferation of artificial intelligence (AI), cloud computing, and 5G networks, the demand for massive data throughput in optical interconnects has grown exponentially [1–3]. This necessitates optical transceivers and engines with broader operational bandwidth, lower power consumption, and higher reliability and scalability for mass production [4,5]. Silicon photonics (SiPh) has emerged as a preeminent platform to meet these challenges, leveraging its inherent compatibility with complementary metal-oxide-semiconductor (CMOS) processes to enable cost-effective, high-volume manufacturing [6–8]. The silicon modulator, serving as a foundational photonic building block within silicon photonics, is tasked with the conversion of electrical signals into optical signals. Within this ecosystem, the Mach–Zehnder modulator (MZM) remains a cornerstone due to its superior wavelength agility and environmental robustness compared to resonant structures [9–12]. However, silicon MZMs are inherently constrained by the plasma dispersion effect [13], resulting in a fundamental trade-off between modulation efficiency and operational bandwidth [14]. In addition, the millimeter-scale footprint of MZMs, significantly larger than that of MRMs, constrains their deployment in high-density interconnect of scale-up networks. While heterogeneous integration with materials like thin-film lithium niobate [15–19] or plasmonic polymers [20–23] offers high performance, these approaches often introduce fabrication complexities that hinder large-scale commercialization. To overcome these limitations in a silicon photonics platform, slow-light enhancement via photonic crystal (PhC) waveguides has been proposed as a promising paradigm [24–28]. By leveraging the slow-light effect, the group velocity of light can be increased and the light–matter interaction can be enhanced, allowing for a drastic reduction in device footprint without sacrificing modulation depth [29–31]. To date, high-performance slow-light modulators have primarily been demonstrated in the C-band for long-haul transmission [32]. In contrast, short-reach interconnects (typically < 10 km)—such as data center networks, co-packaging optics (CPO), and 5G fronthaul—predominantly operate in the O-band to exploit the zero-dispersion window of optical fibers. Developing high-performance O-band slow-light MZMs remains challenging because smaller feature sizes required for O-band PhC structures are highly sensitive to optical proximity effects (OPEs) and critical dimension (CD) variations during fabrication [33,34]. Such sensitivity has long impeded the transition of slow-light devices from laboratory prototypes to stable, wafer-scale production.

In this study, we present a manufacturable, miniaturized O-band slow-light MZM fabricated on a proprietary 300 mm silicon photonics platform. By leveraging a 40 nm process node and optic-electronic co-design methodology, we precisely control the PhC bandgap and p–n junction capacitance. The modulator has a compact footprint of 500 μm and achieves a low $V_\pi \cdot L$ of 0.65 V·cm together with a median EOBW of 94.7 GHz. Without complex AI-based equalizers, we experimentally demonstrate high-speed transmission rates of 300 Gbps/λ (PAM-4) and 400Gbps/λl (PAM-8). To our knowledge, this is the first demonstration of O-band slow-light modulators produced on a 300 mm platform, paving the way for next-generation high-density scale-out optical interconnects.

2. DESIGN AND PRINCIPLES

The architecture of our miniaturized O-band MZM is illustrated in Fig. 1(a). The slow-light PhC waveguides in the modulation arms are designed as a series of Bragg gratings. The 1D PhC waveguide is implemented on a standard 220 nm SOI platform. Each unit cell is defined by two finger-like features extending along the x direction, with a lattice constant Λ of 245 nm to align the slow-light regime with the O-band window [Fig. 1(b)]. The photonic bandgaps can be tuned by adjusting the lattice constant Λ of the unit cell. Strong dispersion occurs near the Brillouin zone (BZ) within the bandgap. Furthermore, the periodicity induces a degeneracy of slow-light modes at the BZ edge, a condition enforced by the glide-plane symmetry of the interface. By setting the geometric parameters ($W_1 = 0.8\Lambda$, $W_2 = 0.7\Lambda$, and $W_3 = 0.45\Lambda$), we induce a degeneracy of slow-light modes at the BZ edge.

Figure 1(c) depicts the numerical dispersion characteristics of the PhC waveguide within the irreducible first BZ. The simulation reveals TE-like bands that can be excited by the fundamental TE mode of the single-mode input ridge waveguide. Owing to the geometric symmetry at the ridge-to-PhC interface, the design selectively excites only even-parity bands with respect to the x = 0 plane [31]. Consequently, odd-parity modes, including the first-order mode, are effectively suppressed due to the vanishing overlap integral of their transverse file profiles with the symmetric input mode. The contributions of the fundamental mode (TE mode) and the second-harmonic mode are shown in blue and red in Fig. 1(c), respectively. Using finite-difference time-domain (FDTD) method, the two-dimensional (2D) normalized intensity profiles corresponding to these two excitation modes are shown in Figs. 1(d) and 1(f). To enhance modulation efficiency, the PhC waveguide is designed to operate in the TE mode to implement phase shift in the MZM, because the TE field has strong spatial overlap with the carrier depletion region located at the lateral center of the waveguide.

The group index ($n_g = c_0 \partial\omega/\partial k$, c_0 denotes the speed of light in vacuum) is proportional to the slope of the bands at each value ($k\Lambda/2\pi$) in Fig. 1(c) [35]. Figure 1(e) (blue curve) shows that the simulated n_g for the fundamental TE mode reaches approximately 8.2 at 1320 nm. The plasma dispersion effect enables refractive index modulation in the MZM by varying the carrier concentration in the active region under an applied voltage. This index change directly results in phase modulation, described by [28]

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot L \cdot \Delta n_{\mathrm{eff}} \propto \frac{2\pi}{\lambda} \cdot N\Lambda \cdot n_g, \tag{1}$$

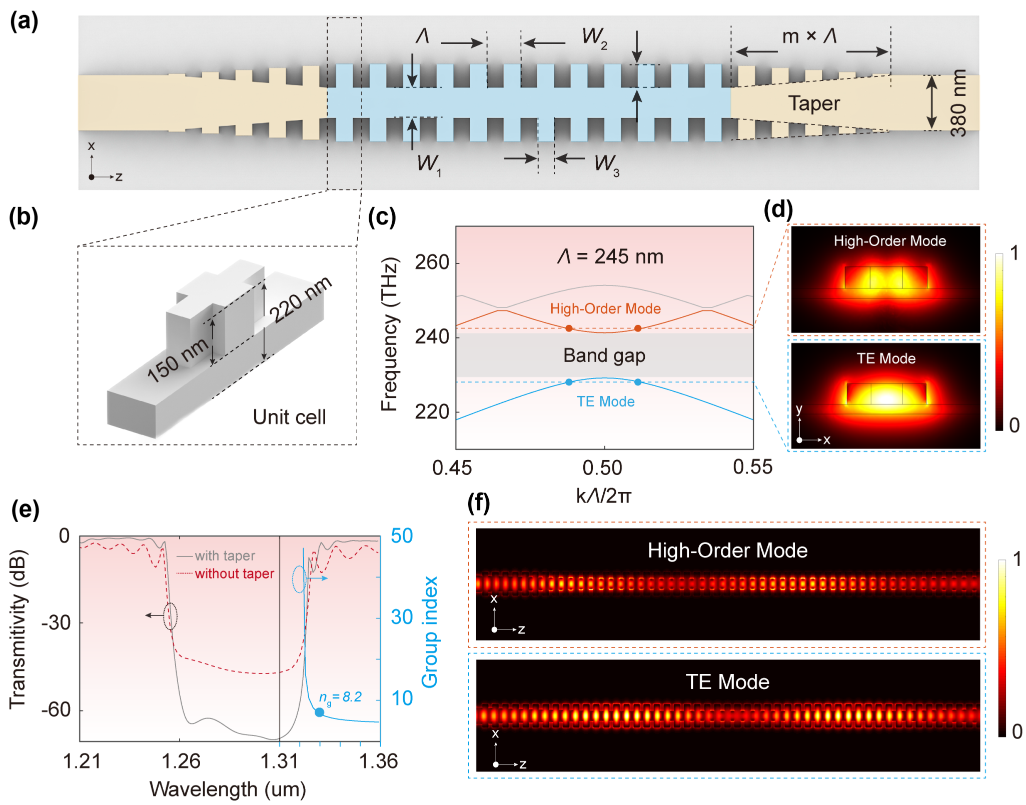

Fig. 1. (a) 3D sketch of the 1D PhC waveguide on an SOI wafer, consisting of a series of fishbone-like Bragg gratings. (b) Unit cell dimensions of the 1D PhC waveguide. (c) Dispersion diagram of the 1D PhC for TE-like polarization. The blue line depicts the contribution of the fundamental mode (TE mode), and the orange line shows the contribution of the high-order mode. The design parameters are $\Lambda = 245$ nm, $W_1 = 0.8\Lambda$, $W_2 = 0.7\Lambda$, and $W_3 = 0.45\Lambda$. (d) Normalized intensity profiles for the fundamental and high-order modes in the x–y plane. (e) Transmission spectra of the 1D PhC waveguide with and without taper sections. The blue curve represents the simulated n_g of the PhC unit cell. (f) Simulated normalized intensity profiles of the high-order mode and the TE mode.

where λ denotes the wavelength, Δn_{eff} is the change in the effective refractive index per unit length in the slow-light waveguide under an applied bias voltage, and N is the number of grating periods. Near the photonic bandgap, the rapidly increasing n_g strengthens optical confinement in the PhC waveguide, extending the photon lifetime and consequently improving modulation efficiency [36]. However, loss mechanisms such as material absorption and scattering limit the usable values of n_g. We therefore chose a moderate n_g of ∼8 to balance modulation efficiency and propagation loss. To reduce mode-field mismatch between the Bloch modes of the PhC and the evanescent Gaussian fields of the ridge waveguides, adiabatic tapers were incorporated at both ends of the 1D PhC section [Fig. 1(a)]. Linear transitions of geometric parameters W_1 and W_2 over 20 periods effectively suppressed Fresnel reflections and coupling losses. Figure 1(e) compares the simulated transmission spectra with and without the tapers, highlighting their critical roles. More design details about the linear transitions are provided in Section 1 of Supplement 1.

The EOBW of the MZM is further analyzed using the frequency response equation [30]:

$$S(\boldsymbol{\omega}) = \left|\frac{2Z_{\mathrm{in}}}{Z_{\mathrm{in}} + Z_s}\right| \left|\frac{(Z_t + Z_0)F_+ + (Z_t - Z_0)F_-}{(Z_t + Z_0)e^{\boldsymbol{\gamma}_0 L} + (Z_t - Z_0)e^{-\boldsymbol{\gamma}_0 L}}\right|, \quad (2)$$

where ω is the microwave angular frequency, Z_s is the source impedance, Z_t is the termination resistance of the modulator, Z_0 is the characteristic impedance of the modulator, Z_{in} is the RF line input impedance, L is the length of the electrode, and γ_0 is the microwave propagation constant. $F_\pm$ are forward/backward propagating waves. Z_{in} and $F_\pm$ can be expressed as

$$F_\pm = \frac{1 - e^{\pm\boldsymbol{\gamma}_0 - j\frac{\boldsymbol{\omega}}{c_0} n_g L}}{\pm\boldsymbol{\gamma}_0 - j\frac{\boldsymbol{\omega}}{c_0} n_g L},$$

$$Z_{\mathrm{in}} = Z_0 \frac{Z_t + Z_0 \tanh(\boldsymbol{\gamma}_0 \cdot L)}{Z_0 + Z_t \tanh(\boldsymbol{\gamma}_0 \cdot L)}. \quad (3)$$

Under impedance matching conditions, Eqs. (2) and (3) indicate that the EOBW of an MZM is predominantly limited by the microwave propagation constant γ_0. As a result, the primary motivation for our PhC MZM design is to minimize the microwave propagation loss by carefully engineering the junction capacitance (C_{pn}) and series resistance (R_{pn}).

The junction capacitance C_{pn} is a key parameter that dictates the trade-off between modulation efficiency and EOBW. While higher C_{pn} strengthens the plasma dispersion effect and improves modulation efficiency, it also reduces junction bandwidth and increases microwave loss, which directly degrades the EOBW. We calculated the EOBW and half-wave voltage as functions of C_{pn} and phase-shifter length (L), as illustrated in Figs. 2(a) and 2(b). The analysis indicates that an EOBW of ∼100 GHz requires $C_{\mathrm{pn}} < 3.3$ pF/cm and a phase shifter length shorter than 700 μm. To balance EOBW, modulation efficiency, and waveguide loss, the slow-light modulator was designed with a length of 500 μm, a C_{pn} of approximately 3.0 pF/cm, and a V_π of about 13.0 V, as indicated by the star markers in Figs. 2(a) and 2(b).

To realize the target C_{pn} in the slow-light modulator, a three-level gradient doping profile was applied, as depicted in Fig. 2(c). Light-doping (N and P) enables the embedding of the phase shifter's p–n junction. Intermediate doping (N+ and P+) serves as concentration-gradient transition to reduce the series resistance R, while N++ and P++ regions provide ohmic contact with the metal layer. A doping dose sweep was performed in the lightly doped region to obtain the optimal C_{pn}. Technology computer-aided design (TCAD)–Lumerical co-simulation was used to analyze the junction capacitance. Figure 2(d) depicts the C_{pn} under 0-bias voltage for different light-doping levels, where C_{pn} increases as the doping dose rises from low to high. A mid-doping dose was chosen to achieve the target capacitance. Based on doping dose and annealing conditions, we simulated the doping concentration profile of the PhC waveguide unit cell. Figure 2(f) illustrates the simulated 2D doping concentration profile. The lateral p–n junctions have light-doping concentrations of 3.8×10^{17} cm^{-3} (P) and 7.1×10^{17} cm^{-3} (N). The intermediate-doping region, located 400 nm from the p–n junction, has a concentration of 2.8×10^{18} cm^{-3}. The P++ and N++ regions are doped to 1.9×10^{20} cm^{-3}.

The frequency- and bias-dependent behavior of C_{pn} is further characterized in Fig. 2(e). At a fixed small-signal frequency, C_{pn} is inversely proportional to bias voltage. Applying a high reverse bias to the MZM depletes the active region by sweeping out electrons and holes from their respective doping regions. The expanded depletion region reduces charge storage, thus decreasing the p–n junction capacitance. Under a reverse bias of −3 V and a modulation frequency of 100 GHz, the depletion region expands significantly, reducing the junction capacitance to ∼2.3 pF/cm, consistent with the design value. Additional simulation data are provided in Section 2 of Supplement 1.

Differential drive modulators effectively suppress common-mode noise and improve linearity, enabling high-order modulation formats such as PAM-4/8. Since differential signaling is widely used in high-speed data interfaces (e.g., SerDes, USB, and PCIe), silicon photonic modulators with differential drivers offer better compatibility. The microwave architecture of the slow-light modulator employs a ground-signal-signal-ground (GSSG) differential electrode configuration [Fig. 2(g)]. By cascading two p–n junctions in a series push–pull configuration, the effective capacitance is halved, thereby increasing the modulator EOBW. To ensure optimal gain and linearity, a thermal phase shifter is integrated on one arm of the modulator to precisely control the quadrature bias point ($P_\pi = 31.7$ mw; details about the thermal phase shifter are provided in Section 9 of Supplement 1). Signal integrity analysis of the GSSG electrodes was performed in ANSYS Electronics, and an optoelectrical co-simulation model was constructed with TCAD and Lumerical. The electrode parameters ($W_g = 100$ μm, $W_{gs} = 25$ μm, $W_s = 15$ μm, and $W_{ss} = 30$ μm) were optimized to improve device performance. As shown in Figs. 2(h) and 2(i), the loaded GSSG electrodes exhibit a characteristic impedance Z_0 of approximately 72 Ω at −3 V bias. This impedance-tailored design, combined with the compact 500 μm slow-light section, effectively mitigates the parasitic microwave loss and reflection, enabling a high EOBW of ∼100 GHz. Figures 2(h) and 2(i) present the calculated microwave loss and characteristic impedance for the GSSG electrodes loaded with a p–n junction. Compared to the unloaded case, the loaded p–n junction decreases the characteristic impedance and increases the loss constant. (More details about the signal integrity analysis are provided in Section 3 of Supplement 1.) To improve impedance matching under bias voltage, the termination resistance of the modulator was set to $Z_t = 65$ Ω.

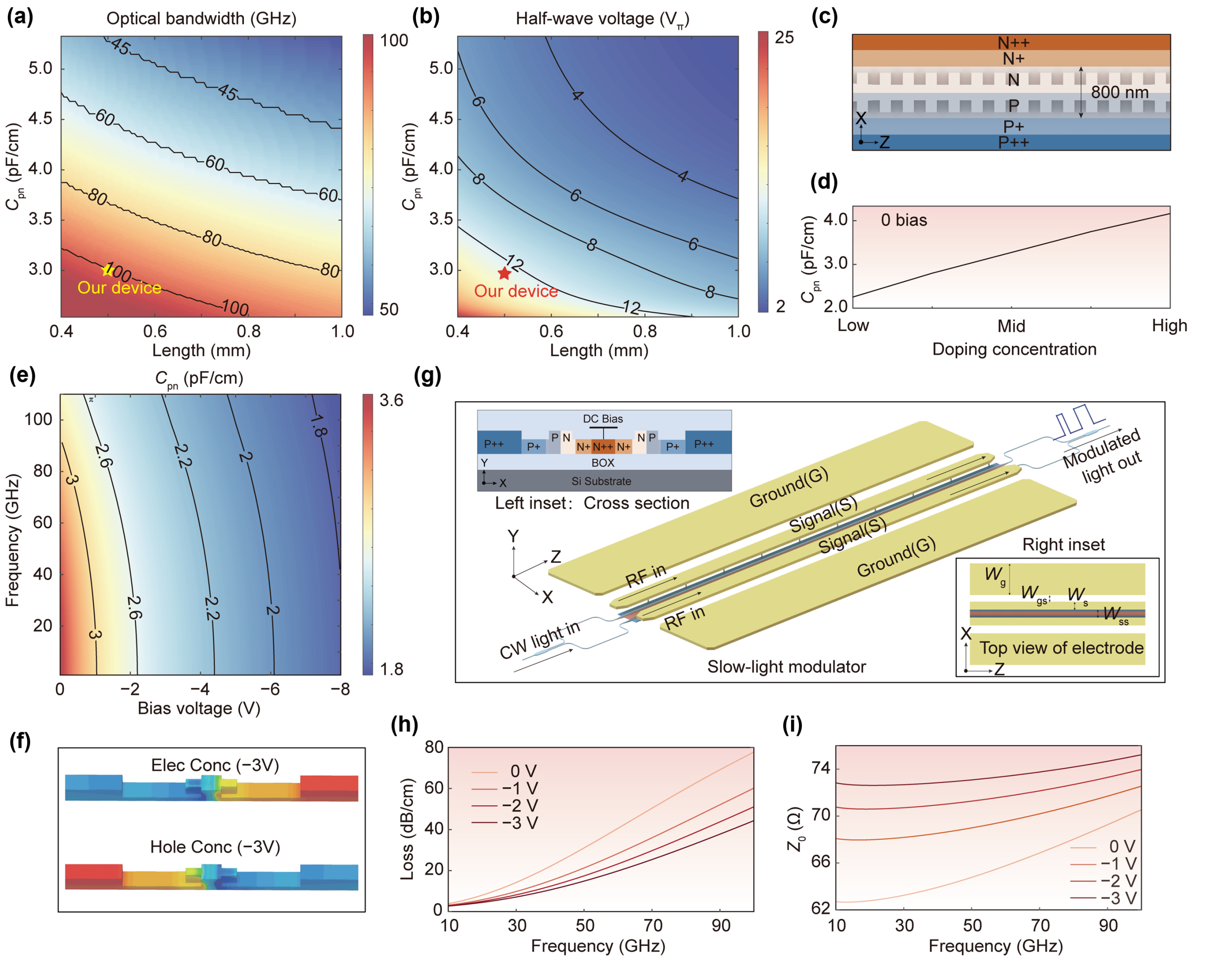

Fig. 2. Design of the slow-light MZM with customized doping and lumped electrodes. (a), (b) Simulation EOBW and half-wave voltage as functions of C_{pn} and phase shifter length. (c) Gradient-doping configuration of the slow-light modulator. (d) Variation of C_{pn} as the light-doping concentration increases from low to high levels. (e) Simulation C_{pn} as a function of bias voltage and small-signal frequencies. (f) TCAD simulation of the doping concentration profile in a 1D PhC waveguide unit cell. (g) Schematic configuration of the proposed slow-light modulator. Left inset: cross-sectional schematic of the slow-light modulator. Right inset: top view of the electrode structure. (h) Microwave loss of the lumped electrode loaded with p–n junction under different bias voltages. (i) Characteristic impedance of the loaded electrode under different bias voltages.

3. DEVICE FABRICATION AND CHARACTERIZATION

The proposed slow-light modulator was fabricated on a 300 mm SiPh platform with a 40 nm process node. The advanced lithography and etching capability ensure the high-performance manufacturing of the PhC waveguides and wafer-scale production of the device. Figure 3 shows the photograph of a fully processed 300 mm SOI wafer and the slow-light MZM.

Wafer-level characterization was conducted to evaluate insertion loss (IL) and modulation efficiency. Figure 4(a) shows the static transmission spectra, showing a measured insertion loss (IL) of 2.9 dB. The blue-shaded region in the figure marks the fast-light region, which is located away from the BZ edge. As the wavelength approaches the BZ edge (highlighted in red), the slow-light effect becomes prominent, with a wavelength-dependent group index n_g and V_π. To balance the enhancement of n_g against the increasing insertion loss near the bandgap, 1319 nm (marked by a solid red dot) was chosen as the optimal operating wavelength, and the measured V_π is 13.1 V [Fig. 4(b)], corresponding to a record-low

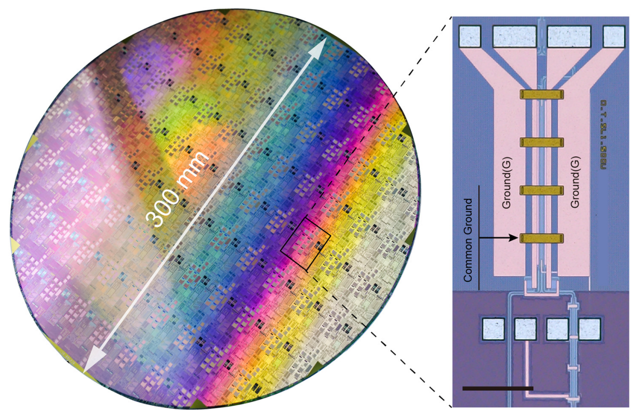

Fig. 3. Optical images of a fabricated 300 mm SOI wafer and a slow-light modulator, scale bar: 200 μm.

$V_\pi \cdot L$ of 0.65 V · cm for a 500 μm interaction length. Across the entire wafer, the median IL in the fast-light region is 2.4 dB (excluding grating couplers). The maximum and minimum values

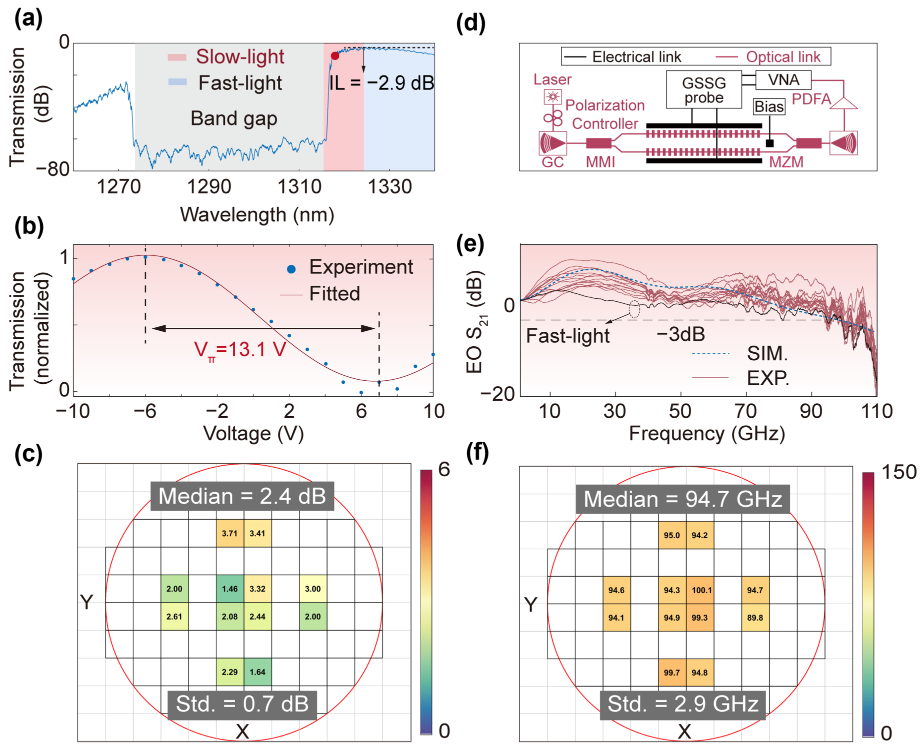

Fig. 4. Device characterization and static EO performance. (a) The transmission spectrum of the slow-light MZM. (b) Normalized optical transmissions of the fabricated slow-light modulator as a function of the applied voltage, showing a V_π of 13.1 V. (c) Wafer-level measurement results for IL in the non-slow-light region. (d) Experimental setup for EOBW measurement. (e) and (f) Measured EO S_{21} curves and 3 dB EOBW at different wafer locations, with a median EOBW of 94.7 GHz and a standard deviation of 2.9 GHz.

are 3.71 and 1.46 dB, respectively, with a standard deviation of 0.7 dB, demonstrating excellent process uniformity [Fig. 4(c)]. Detailed passive characterization of the slow-light waveguides is provided in Sections 4 and 5 of the Supplement 1.

The EOBW was characterized using a high-frequency small-signal analysis setup, as shown in Fig. 4(d). The measured EO S_{21} curves from multiple dies on the 300 mm wafer are plotted in Figs. 4(e) and 4(f), revealing a median 3 dB EOBW of 94.7 GHz, with a maximum of 100.1 GHz, a minimum of 89.8 GHz, and a standard deviation of 2.9 GHz. The experimental results agree well with TCAD–Lumerical co-simulations [dashed blue line in Fig. 4(e)], validating the p–n junction engineering and microwave-optical phase matching. The uniformity across the 300 mm wafer highlights the potential of slow-light modulators to move from laboratory prototypes to mass-producible components. A representative die from the wafer center was selected for a direct performance comparison with the slow-light operation. The EOBW of the same modulator was measured in the fast-light region (away from the band edge). As shown by the black solid line in Fig. 4(e), the measured 3 dB EOBW in the fast-light region is 73 GHz, lower than the 94.7 GHz achieved in the slow-light region. This reduction is mainly attributed to weaker light–matter interaction away from the BZ edge, consistent with the reduced group index and modulation efficiency in the fast-light regime. Additional comparative measurement results are provided in Section 8 of Supplement 1.

To evaluate the modulator in high-capacity interconnect scenarios, back-to-back (BtB) transmission experiments were conducted using NRZ, PAM-4, and PAM-8 formats [Fig. 5(a)]. For NRZ signaling, clear eye openings were obtained for 56 Gbaud and 112 Gbaud PRBS NRZ, with extinction ratios (ERs) of 4.3 and 3.2 dB, respectively, as shown in Figs. 5(b) and 5(c). In addition, the device also performs well with advanced modulation formats. Driven by 112 and 150 Gbaud PAM-4 SSPRQ patterns, the modulator achieved transmitter dispersion eye closure quaternary (TDECQ) values of 1.05 and 1.8 dB, respectively, while maintaining a symbol error rate (SER) below 2E-2 threshold, as illustrated in Figs. 5(d) and 5(e). We also demonstrated 400 Gbps/λ transmission using 135 Gbaud PAM-8 modulation, confirming the high-data-rate operation of the modulator, as shown in Fig. 5(f). In addition, the 400 Gbps PAM-4 transmission capability of the slow-light modulator is analyzed in Section 10 of Supplement 1. Figure 5(g) presents the BER measured for different PAM-4 signal rates. Based on the soft decision forward error correction (SD-FEC) threshold of 2.0×10^{-2}, the sensitivity of 160, 220, and 280 Gbps PAM-4 signals is determined under different received optical power of 7, 5, and 3 dBm, respectively. The measured BER is likely conservative, since the photodetector and the probe used in this setup have limited bandwidth.

In the proposed device, electrical power is dissipated during rising transitions as the junction capacitance C_{pn} is charged. The junction capacitance is estimated to be 2.3 pF/cm under a −3 V bias. For the 500 μm long slow-light modulator with a p–n junction series configuration, the junction capacitance C_j is approximately 57.5 fF. With a peak-to-peak drive voltage swing of 2.7 V_{pp}, the power consumption is about 15.0 fJ/bit. Further details can be found in Section 7 of Supplement 1.

Table 1 compares our device with state-of-the-art slow-light modulators. Operating in the O-band, our device demonstrates a median EOBW of 94.7 GHz, a $V_\pi \cdot L$ of 0.65 V · cm, and an

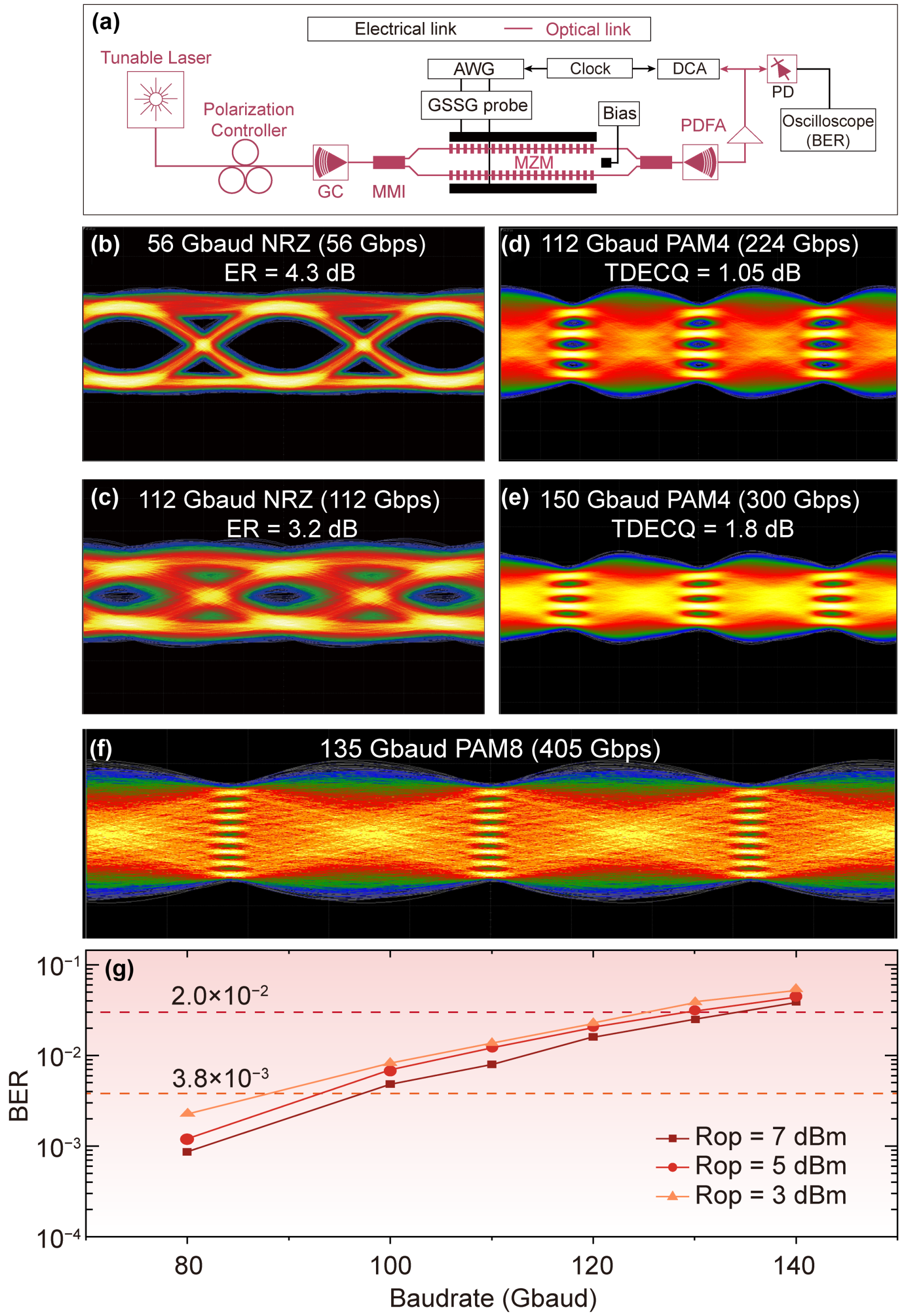

Fig. 5. (a) Schematic diagram of the experimental setup for optical transmission and BER experiment. Eye diagrams for NRZ at data rates of (b) 56 Gbps and (c) 112 Gbps. Measured PAM-4 eye diagrams at (d) 224 Gbps and 300 Gbps. (f) Measured PAM-8 eye diagrams at 405 Gbps. (g) Measured BER for the PAM-4 under different received optical powers.

Table 1. Performance of Various State-of-the-Art Slow-Light Modulators

Platform	Operating Wavelength	$V_\pi \cdot L$	EOBW	Arm Length	IL	V_{pp}	Data Rate	Ref.
Si	O-band (TW-MZM)	1.1 V cm	44 GHz	3000 μm	4.6 dB	2.3 V	336 Gbps	[37]
Si	C-band	0.96 V cm	110 GHz	124 μm	6.8 dB	5.0 V	116 Gbps	[28]
Si	C-band	0.32 V cm	31 GHz	/	8.0 dB	0.87 V	64 Gbps	[27]
Si	C-band	0.82 V cm	90 GHz	249 μm	10.5 dB	5.0 V	400 Gbps	[29]
Si	C-band	/	110 GHz	10 μm	2.8 dB	2.0 V	100 Gbps	[32]
Si	C-band	0.51 V cm	42 GHz	570 μm	5.5 dB	7 V	100 Gbps	[38]
Si	C-band	0.53 V cm	85 GHz	201 μm	5.3 dB	0.8 V	128 Gbps	[39]
This work	O-band	0.65 V cm	94.7 GHz	500 μm	2.4 dB	2.7 V	400 Gbps	

IL of only 2.4 dB over a 500 µm long arm, enabling a data rate of 400 Gbps under a 2.7 V_{pp} differential drive. Compared with reported silicon slow-light modulators, the proposed slow-light MZM fabricated on a 300 mm SiPh platform offers a competitive combination of high bandwidth, low loss, high modulation efficiency, and excellent fabrication uniformity, making it a promising candidate for next-generation scale-out and scale-up optical networks.

4. CONCLUSION

In this work, we have demonstrated high-performance O-band slow-light MZMs fabricated on 300 mm silicon photonic platform. By leveraging the slow-light effect near the first BZ edge in 1D PhC waveguides, these modulators achieve a compatibility between compact footprint and high modulation efficiency, while also maintaining high EOBWs.

A key technology in our work is the so-called design-process co-optimization for photonic integrated circuits. By establishing a rigorous analytical model of the p–n junction capacitance and its impact on microwave propagation, we achieved a near-ideal balance between carrier-induced phase shift and RC-limited bandwidth. The customerized three-level gradient doping, enabled by the precision of the 40 nm CMOS node, allowed us to integrate longer phase shifters without compromising speed. This design ensures a higher dynamic ER and a lower required drive voltage, which are essential for supporting energy-efficient PAM-4/8 signaling. Our device supports raw data rates up to 400 Gbps (135 Gbaud PAM-8) without the reliance on power-hungry AI-based digital signal processing (DSP) or complex equalizers, which is particularly important for short-reach intra-datacenter links where the near-zero dispersion window of the O-band minimizes pulse broadening and simplifies the transceiver architecture. The transition from C-band to O-band slow-light devices represents a significant engineering step, while the feature sizes of O-band PhC structures impose stringent demands on lithographic fidelity and etch uniformity. To the best of our knowledge, this work represents the first demonstration of high-performance O-band slow-light modulators on SiPh platform.

The design method presented in our work provides a versatile basis for developing modulators that facilitate multi-dimensional multiplexing, including wavelength-, polarization-, and mode-division multiplexing. Such modulators could substantially increase the aggregate transmission capacity and improve optical interconnect performance for next-generation optical communication and AI-driven computing.

5. METHODS

A. Fabrication of the Devices

The slow-light modulator was fabricated on a 300 mm high-resistance SOI wafer, featuring a 220 nm thick silicon layer and a 2 µm thick buried oxide (BOX) layer, using a proprietary standard 40 nm lithography SOI process. The active device fabrication comprised five ion implantation stages, three copper interconnect layers, and a single aluminum electrode. To ensure an adiabatic transition of the waveguide mode into the slow-light mode, waveguide tapers are employed to connect the single-mode waveguide (380 nm) with the phase-shifting region.

B. Experimental Details

For the bandwidth test, an O-band tunable laser source was coupled in and out of the modulators using a pair of grating couplers. The polarization states of light are aligned using a polarization controller. The praseodymium-doped fiber amplifier (PDFA, Thorlabs PDFA100) is connected after the MZM output to compensate for optical loss. After high-speed OE conversion by a photodetector (PD), the modulated optical signal was fed into a 110 GHz bandwidth lightwave component analyzer (LCA, Keysight N4372E). The EO response of the device was characterized by driving the modulator with a sinusoidal signal generated by the LCA, applied via a high-speed GSSG microwave probe. To apply both RF and DC bias signals, a 110 GHz bias tee was employed. Systematic errors, specifically the excess losses introduced by RF cables and connectors, were de-embedded from the measurements.

In the data transmission experiments, the RF signal (2.7 V_{pp}) generated by a 224 GSa/s arbitrary waveform generator (AWG, Keysight M8199B) was applied to the GSSG RF probe. The AWG provided PRBS11 NRZ and SSPRQ PAM-4 signals to drive the modulator. A 16 GHz reference clock, generated by a clock source (Keysight M8008A), was used to trigger both the AWG and a digital communication analyzer (DCA, Keysight, N1000A). The output optical signal from the modulator was sent through a PDFA and detected by the DCA. A fourth-order Bessel filter and a feed-forward equalizer (FFE) were employed at the receiver to mitigate noise, without implementing advanced artificial intelligence (AI) algorithms such as a deep-learning neural network (DNN). Bit error ratio measurements used the same configuration, with a Keysight M8199B arbitrary waveform generator providing the programmable signals. The received signals were then fast-sampled using a real-time oscilloscope (Keysight UXR0594AP) and subsequently processed offline by applying feed-forward equalization (FFE) algorithms.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are available from the corresponding author upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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